RWAXI Platform

Architecture design: C4 diagrams modelled in Archi

Stellar Community Fund #46 - Build Award, Open Track

Prepared by Nekot (technical provider) for RWAXI Corp.

Document status	Draft for team review
Source model	`docs/architecture/archi/rwaxi-architecture.archimate` (Archi / ArchiMate)
AI disclosure	The model and this document were drafted with AI assistance (Claude, Anthropic) from the team's technical brief and the architecture documents in the repository. They must be reviewed and adjusted by a named team member before submission. Logged in `AI_DISCLOSURE.md`.
Images	The figures in this draft are previews rendered from the model. Final figures are to be exported from Archi.

1. Global objective

RWAXI issues and administers **regulated real-world-asset (RWA) security tokens** on Stellar. The first vehicle, Avista Beach Tokenization Fund I, tokenizes villas and lots in Los Córdobas, Colombia, distributed under US Regulation S. The architecture has one central claim:

> **Transfer restrictions live inside the token contract, not in an off-chain database.** Every transfer, mint and burn runs an identity check and a set of compliance hooks in the same transaction; if any rule fails, the whole operation reverts. No operator database, API or third-party wallet can bypass it.

The design is built to serve the SCF Build Award criteria:

SCF criterion	How the architecture answers it	Where
Clear on-chain integration plan specific to Stellar	OpenZeppelin *stellar-contracts* RWA token, identity and compliance modules; OZ smart accounts with passkeys; OZ Relayer / Channels for fees; USDC through the Stellar Asset Contract; SEP-24 anchor for COP	Views 2, 3
Technically ambitious, clearly scoped	Only a few custom contracts (Reg S module, subscription, distribution, checkpoint, factory); everything else is OZ code, configured	View 3
Uses the network's latest strengths	Soroban SDK 27, OZ stellar-contracts 0.7.1, smart accounts, passkey signing, sponsored fees	Views 2, 3
Milestones mapped to mainnet	Custom components are marked by tranche (T2: subscription, distribution, checkpoint); testnet before mainnet	View 3
Measurable on-chain growth	An event indexer reads contract events into a database, so activity is measurable and publicly verifiable on Stellar Expert	Views 1, 2

2. How to read the diagrams

The three views follow the C4 model, from the widest to the most detailed level, modelled in ArchiMate so they can be maintained in Archi:

- **Level 1 - System context:** who uses RWAXI and which external systems it depends on.
- **Level 2 - Containers:** the deployable pieces, off-chain and on-chain, and how they talk.
- **Level 3 - Components (on-chain):** the smart contracts and the compliance logic.

Colour code: **blue** = OpenZeppelin code (configured, not rewritten); **orange** = RWAXI custom code; **grey** = external system or asset; **yellow** = people; **green outlines** = groupings. Numbers on arrows give the order of the main flow. Arrows are flows (solid) or associations (dashed).

3. View 1 - System context (C4 level 1)

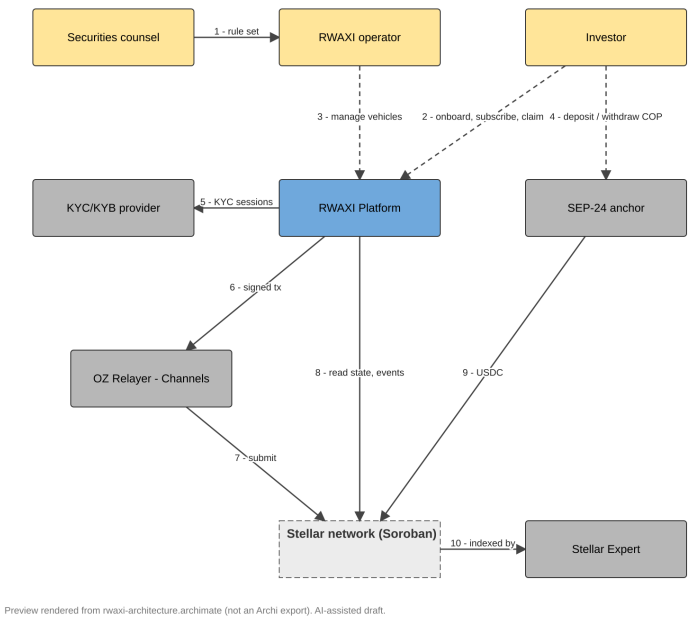


Figure 1. System context. Preview rendered from the Archi model.

Purpose. Shows RWAXI as one system and everything that surrounds it, so a reviewer can judge the scope quickly.

Element	Role
Investor	Colombian or international. Creates a passkey account, passes KYC, subscribes in USDC, claims distributions.
RWAXI operator	Creates vehicles, approves investors, mints and pays out through the backoffice.
Securities counsel	Defines transfer rules per vehicle. Never writes to the system directly; the operator configures the modules.
KYC/KYB provider	Regulated identity verification (documents, liveness, AML).
OZ Relayer - Channels	Submits signed transactions and pays fees.
SEP-24 anchor	Licensed fiat rails, COP to and from USDC.
Stellar network (Soroban)	Settlement layer: token, identity and compliance contracts, USDC.
Stellar Expert	Public verification of every deployed contract and transaction.

What it shows on purpose. RWAXI holds no investor keys (passkeys stay in the device's secure enclave), handles no fiat (only a licensed anchor does) and stores no personal data on-chain. Each external dependency has a documented failure impact and fallback: a second RPC provider, a provider-abstraction layer for KYC, a self-hosted OZ Relayer with the same API, and direct USDC deposits if the anchor is down.

4. View 2 - Containers (C4 level 2)

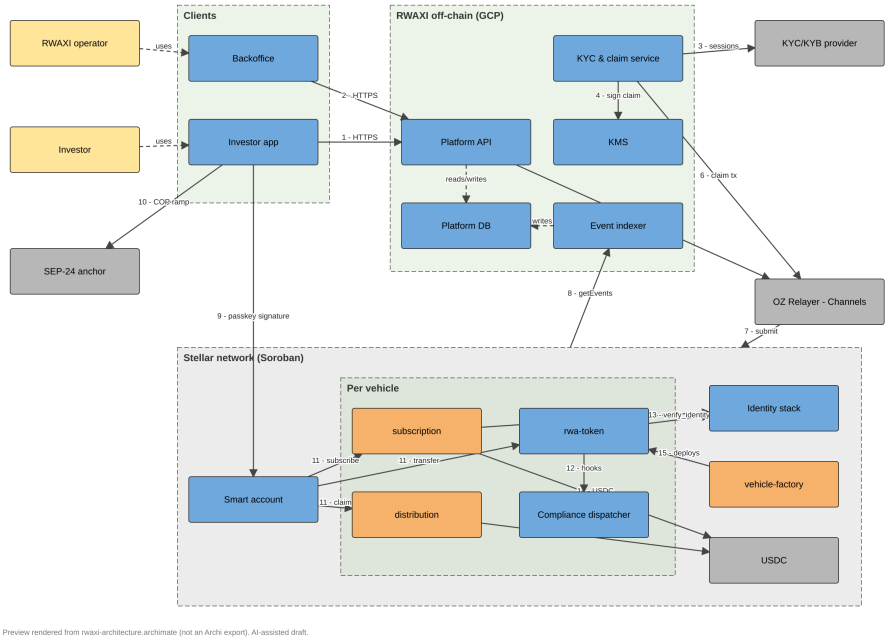


Figure 2. Containers. Preview rendered from the Archi model.

Purpose. Shows every deployable piece and the trust boundary between the off-chain services and the contracts.

Group	Containers	Key rule
Clients	Investor app (React Native / Expo); Backoffice (Next.js)	The app signs with the investor's passkey. The backoffice uses SSO plus a hardware key; mainnet actions go through a multisig.
Off-chain (GCP)	Platform API (NestJS); KYC and claim service; Event indexer; PostgreSQL; KMS	The API builds and simulates transactions and never holds keys. Only the KYC service can call the KMS key that signs claims. No KYC documents are stored in the database.
Stellar (Soroban)	Smart account; per-vehicle rwa-token, compliance, subscription, distribution; shared identity stack and vehicle-factory; USDC	Every transfer, mint or burn calls the identity verifier and compliance, and fails atomically.

Main flows. (1-2) Clients call the API over HTTPS. (3-4) The KYC service opens sessions with the provider and signs claims in the KMS. (5-7) Signed transactions go through the Relayer to the network. (8) The indexer reads events into the database. (9-11) The app authorizes with a passkey signature on the investor's smart account, which calls the vehicle contracts. (12-13) The token calls compliance hooks and the identity verifier. (14) Subscriptions and distributions settle in USDC. (15) The factory deploys each vehicle's contract set.

5. View 3 - On-chain components (C4 level 3)

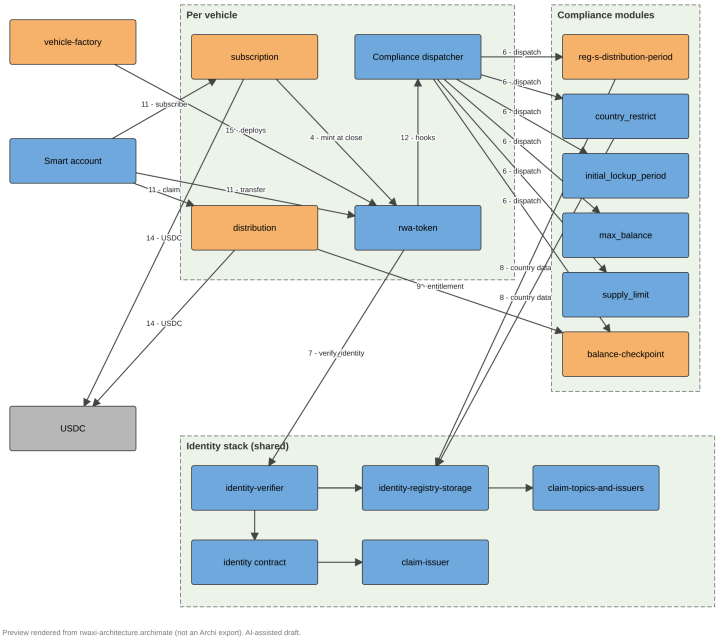


Figure 3. On-chain components. Preview rendered from the Archi model.

Purpose. Shows the compliance model itself, which is the core technical claim of the application.

Component	Source	What it does
rwa-token	OZ RWAToken	Thin wrapper; SEP-41 surface. Checks pause, freezes and free balance, verifies both identities, then runs compliance hooks.
Compliance dispatcher	OZ Compliance	Registers modules per hook (Transferred, Created, Destroyed). Only the module operator can change them; every change emits a monitored event.
reg-s-distribution-period	**Custom**	Until the period's end ledger, no tokens may reach a wallet whose identity has a US country relation.
country_restrict, initial_lockup_period, max_balance, supply_limit	OZ	Sanctioned-country block, holding period, per-identity cap, issuance cap.
balance-checkpoint	**Custom** (T2)	Records per-holder balance per distribution epoch.
subscription	**Custom** (T2)	USDC escrow with soft and hard caps; mints at close or refunds.
distribution	**Custom** (T2)	Pull-model USDC payouts pro rata to the checkpoint.
Identity stack	OZ	identity-verifier, identity-registry-storage (wallet to identity and country), claim-topics-and-issuers, one identity contract per investor, claim-issuer (key in KMS).
vehicle-factory	**Custom**	Deploys the contract set for a new vehicle with deterministic salts.
Smart account	OZ stellar-accounts	One per investor; passkey signer.

Why it matters. A compliance module rejects by panicking, which reverts the whole transaction. Each rule maps to a module and a test (legal rule to module to parameter to test), so no rule is implemented without evidence. The critical flow is a US-resident recipient blocked during the Reg S period, followed by an allowed transfer to a Colombian recipient; both produce public transaction hashes on Stellar Expert.

6. Method and disclosure

The diagrams are maintained as an Archi model (`rwaxi-architecture.archimate`), so elements, relations and views stay consistent and editable by the team. Sequence diagrams (compliance-checked transfer, onboarding, subscription, distribution, device recovery) are kept separately, because Archi does not model sequences natively.

AI use. The Archi model and this document were first drafted with an AI assistant (Claude, Anthropic) from the team's technical brief and the repository's architecture documents. A named team member reviews and adjusts the model in Archi and signs off before submission. Each AI-assisted artifact is listed in `AI_DISCLOSURE.md` with the tool, what it did, who reviewed it and what was checked.

Review pending: reviewer name, date and changes made in Archi to be completed by the technical lead.